SSH Report

- SSH(Secure shell):
  It is a is a cryptographic network protocol used to establish a secure encrypted connection between two machines.
  In Git it is used to authenticate with remote repositories (GitHub) instead of entering username and password every time.

- SSH contains a public key on the hosting account and a private key on my machine.
- The work flow of SSH:
  - Git initiates an SSH connection.
  - The server checks your public key.
  - If it matches your private key → access granted.
- Some of its advantages:
  - Very high security
  - Better in daily use
- Disadvantage of it is it may be hard to setup.
- Default location varies depending on the OS:
  - For windows: C:\Users\Username\.ssh
  - For Linux: ~/.ssh
- Setting up SSH:
  1- In git bash : ssh-keygen -t ed25519 -C "email"
  2- Entering the passphrase then enter ==> two files will be created

3- eval "$(ssh-agent -s)"

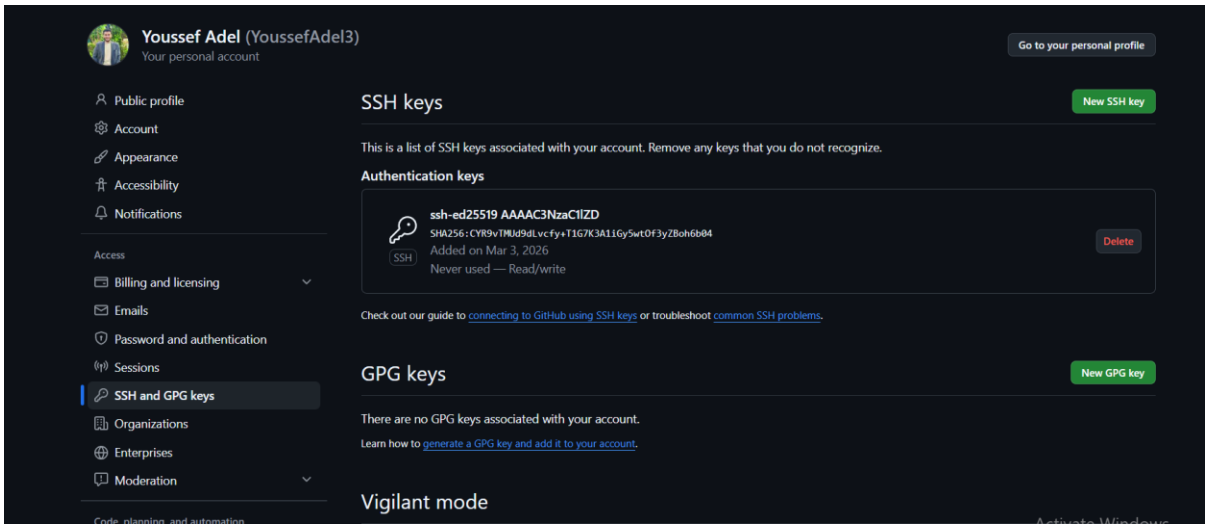
4- ssh-add ~/.ssh/id_ed25519

5- cat ~/.ssh/id_ed25519.pub then copy the result

6- adding ssh key in my github account and paste the key

7- test the key : ssh -T git@github.com it shows authenticated successfully.

- SSH on my account:



- Testing ssh key:

```
you27@Joo MINGW64 ~/OneDrive/Desktop
$ ssh -T git@github.com
The authenticity of host 'github.com (140.82.121.3)' can't be established.
ED25519 key fingerprint is: SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvCOqU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
Hi YoussefAdel3! You've successfully authenticated, but GitHub does not provide
shell access.
```

- Using it while cloning repo:

```
you27@Joo MINGW64 /f/ITI/GitHub/task01
$ git clone git@github.com:ayaatadel/python_r2_2026_mnf.git
Cloning into 'python_r2_2026_mnf'...
Enter passphrase for key '/c/Users/you27/.ssh/id_ed25519':
remote: Enumerating objects: 124, done.
remote: Counting objects: 100% (124/124), done.
remote: Compressing objects: 100% (74/74), done.
remote: Total 124 (delta 31), reused 121 (delta 28), pack-reused 0 (from 0)
Receiving objects: 100% (124/124), 519.59 KiB | 315.00 KiB/s, done.
Resolving deltas: 100% (31/31), done.
```